Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/18/2008
PI Name:	Delp, Michael Ph.D.		
Project Title:	Adaptations of Cerebral Arteries to Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	07/01/2006	**End Date:**	12/31/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI moved to University of Florida from WVU in mid-2007 (12/07) NOTE: Received NCE through 12/31/2007 (original end date was 12/31/2006), per A. Chu (1/07) NOTE: PI moved to West Va. U from TX A&M in Sept 2005 (per A. Chu 7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA06CB42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	These studies address the mechanisms and functional significance of how simulated microgravity affects vascular smooth muscle and endothelial cell function of cerebral resistance arteries. Alterations in vascular function induced by microgravity are particularly relevant to the problem of orthostatic intolerance experienced by astronauts upon re-entry into the Earth's gravitational field. Previous research has demonstrated that reductions in blood volume and altered baroreflex control of the cardiovascular system contribute to, but do not fully account for changes in orthostatic tolerance. To investigate the effects of microgravity on the cerebral circulation, head-down tail-suspension (HDT) of rats were used as a ground-based model of microgravity because HDT, like actual microgravity, induces a headward fluid shift. It is now evident that the shift of fluid toward the head and the corresponding increase in cephalic hydrostatic pressure alter the mechanical forces exerted on cerebral resistance arteries, the vessels responsible for regulating cerebral perfusion. For example, it has now been demonstrated that medial layer (smooth muscle) hypertrophy occurs in cerebral resistance arteries (middle cerebral and basilar) and that vasoconstrictor responsiveness of these arteries to mechanical stimuli (transmural pressure and shear stress) and vasoconstrictor agonist (KCl) is enhanced by HDT. The purpose of the present proposal was to address four specific aims: 1) to determine whether alterations in the release of an endothelium-derived vasodilator substance(s) accounts for the enhanced spontaneous tone and vasoconstrictor responsiveness of the middle cerebral artery, 2) to determine whether the smooth muscle hypertrophy and the thickening of the medial layer alters the mechanical properties (structural behavior and material properties) of the middle cerebral artery, 3) to determine the mechanism(s) of how HDT alters endothelium-mediated vasodilation, and 4) whether reported cerebral vascular alterations affect blood flow distribution and vascular resistance within the brain. This work is justified by the fact that it provides potentially important information about the mechanisms contributing to the orthostatic intolerance of astronauts returning to Earth. Such information may be useful in the development of more effective countermeasures. The impact of simulated microgravity on maximal responsiveness and sensitivity of resistance arteries to a variety of vasoconstrictor and vasodilator agents, each acting through different mechanisms, will be determined. This work will provide potentially important information about the mechanism(s) underlying the orthostatic intolerance of astronauts returning to Earth. Such information would be useful in the development of more effective countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The studies supported through this grant are significant and provide Earth benefits in several respects. First, these studies provide new information regarding how altered physical forces (transmural pressure and shear stress) induced by simulated microgravity impact the biological function and structure of cerebral resistance arteries. Such changes may also occur during prolonged bed rest. Second, blood flow and vascular resistance measurements demonstrate that the effects of cerebral vascular adaptations are not generalized throughout the brain but rather are specific to discrete regions. Such data provide potentially important information about the effects of cerebral vascular remodeling on motor, neurovestibular, sleep/circadian rhythm, and cardiorespiratory control regions within the brain. Third, these studies are significant because they address the problem of reduced orthostatic tolerance in astronauts through an enhanced cerebral autoregulation mechanism. Increased understanding of this vascular mechanism may assist others in the development of more targeted countermeasures to offset the impact of fluid shifts toward the head. And finally, the results from this work may help scientists and clinicians better understand cardiovascular disorders occurring in the general population. For example, orthostatic intolerance is the primary underlying cause of emergency room visits among the elderly, such as hip fracture resulting from light headedness when standing. In addition, endothelial dysfunction occurs with primary aging and underlies many of the cardiovascular pathologies found in western society. Although the stimulus for or mechanism of endothelial dysfunction may vary among the various disorders, results from these studies may provide important new information regarding the impact of cerebral artery endothelial dysfunction on orthostatic intolerance and brain perfusion.
	I. Increases in myogenic, shear stress and agonist-induced vasoconstriction in the middle cerebral artery (MCA) with head-down tail-suspension (HDT) in rats are the result of alterations in the release of the endothelium-derived relaxing factor, nitric oxide (NO). Myogenic, shear stress, and KCl-induced vasoconstriction will be tested in the MCA from control and HDT rats with an intact and removed endothelial cell layer and in the presence and absence of the NO synthase inhibitor NG-nitro-L-arginine methyl ester (L-NAME). These studies have been completed and the results have been published in the American Journal of Physiology: Heart and Circulatory Physiology (Wilkerson, MK, LA Lesniewski, EM Golding, RM Bryan, A Amin, E Wilson, and MD Delp. Simulated microgravity enhances cerebral artery vasoconstriction and vascular resistance through an endothelial nitric oxide mechanism. Am. J. Physiol. Heart Circ. Physiol. 288: H1652-H1661, 2005). The results demonstrate that simulated microgravity enhances basal tone and vasoconstrictor responsiveness of cerebral arteries, therefore potentially limiting cerebral perfusion during periods of altered gravitational stress (e.g., during standing after return from a weightless environment). When applied to the human condition, these results suggest that alterations in cerebral autoregulation associated with microgravity or prolonged bedrest may be the result of enhanced vasoconstrictor tone of cerebral arteries. II. The mechanism through which the endothelium mediates vasodilation of MCAs is altered by HDT, so that there is less vasodilation through the nitric oxide synthase pathway and more vasodilation through the release of endothelium-derived hyperpolarizing factor (EDHF). Vasodilation induced by 2-methylthio-ATP (2MeS-ATP) and bradykinin, two endothelium-dependent vasodilators, will be determined in the presence of the NO synthase inhibitor L-NAME, the cyclooxygenase inhibitor indomethacin, and the Ca2+-activated K+ channel inhibitors charybdotoxin and apamin. These studies have been completed and the results have been published in the Journal of Applied Physiology under a Highlighted Topic in the journal: Regulation of the Cerebral Circulation (Prisby RD, MK Wilkerson, EM Sokoya, RM Bryan, E Wilson, and MD Delp. Endothelium-dependent vasodilation of cerebral arteries is altered with simulated microgravity through nitric oxide synthase and EDHF mechanisms. J. Appl. Physiol. 101: 348-353, 2006). The results demonstrate that 2-MeS-ATP and BK-elicited vasodilation is diminished in MCAs from HDT rats, and that this attenuated endothelium-dependent vasodilation occurs through alterations in the nitric oxide synthase (NOS) signaling pathway. Furthermore, the diminished endothelium-dependent vasodilation is associated with reduced NOS and caveolin-1 protein expression. Concomitant with the attenuated NOS component of the 2-MeS-ATP mediated vasodilation is an enhancement of an EDHF-mediated vasodilator component in MCAs from HDT rats, as determined through the inhibition of Ca2+-activated K+ channels. Although the up-regulation of an EDHF signaling mechanisms appears to partially compensate for the diminished NOS signaling pathway, the net effect is that endothelium-dependent vasodilation in HDT rats remains lower than that in cerebral arteries from control animals. The functional consequence of these vascular alterations with HDT appears to be greater cerebral vascular resistance and lower cerebral perfusion.

Task Progress:	When applied to the human condition, these results suggest that reported alterations in cerebral autoregulation associated with microgravity or prolonged bedrest may be the result of diminished levels of cerebral artery endothelial nitric oxide (NO). Although diminished NO bioavailability may be an appropriate adaptation of the cerebral vasculature to the elevations in arterial pressure associated with chronic headward fluid shifts, this adaptive response may limit the ability of cerebral vasculature to respond to abrupt alterations in pressure, such as that which occurs with orthostatic stress. III. The smooth muscle hypertrophy and the corresponding increase in MCA medial thickness induced by HDT will diminish the passive pressure-diameter relationship and increase the stiffness of MCAs. The structural behavior and material properties of the MCA from control and HDT rats will be compared to determine whether structural vascular alterations affect the stiffness of these arteries, and thus, how readily the arteries will vasodilate during smooth muscle cell relaxation. These studies have been completed and the results have been published in the American Journal of Physiology: Heart and Circulatory Physiology (Wilkerson, MK, LA Lesniewski, EM Golding, RM Bryan, A Amin, E Wilson, and MD Delp. Simulated microgravity enhances cerebral artery vasoconstriction and vascular resistance through an endothelial nitric oxide mechanism. Am. J. Physiol. Heart Circ. Physiol. 288: H1652-H1661, 2005). Results from these studies indicated that although smooth muscle hypertrophy occurred in the smooth muscle (medial) layer of cerebral arteries from HDT rats, it did not change the mechanical properties of the cerebral arteries (i.e., it did not increase stiffness) as indicated by the passive pressure-diameter relation being unaltered. IV. The net effect of HDT-induced alterations in cerebral resistance artery vasoconstrictor responsiveness, vasodilator responsiveness, or mechanical properties is a diminished cerebral perfusion and elevated cerebral vascular resistance. Total cerebral perfusion, brain blood flow distribution, and total and regional cerebral vascular resistance will be determined in control and HDT rats during horizontal (0o tilt) and 75o head-up tilt. These studies have been completed and the results have been published in the American Journal of Physiology: Heart and Circulatory Physiology (Wilkerson, MK, LA Lesniewski, EM Golding, RM Bryan, A Amin, E Wilson, and MD Delp. Simulated microgravity enhances cerebral artery vasoconstriction and vascular resistance through an endothelial nitric oxide mechanism. Am. J. Physiol. Heart Circ. Physiol. 288: H1652-H1661, 2005). The results demonstrate that following a 2 wk period of simulated microgravity, cerebral perfusion is lower and vascular resistance is higher in several regions of the brain when the animals are normally standing in the horizontal position. During head-up tilt, the HDT rats become hypotensive. However, despite this hypotension, cerebral vascular resistance is higher and perfusion is lower than that of control animals in several regions of the brain, including those involved in cardiorespiratory control, the maintenance of equilibrium, hearing, vision and circadian rhythm regulation. Therefore, these data suggest that spaceflight and bedrest-induced deficits in locomotion, balance and orthostatic tolerance may be related to regional reductions in perfusion of key regulatory areas within the brain. V. Additional Studies: Results from Specific Aim IV demonstrated that during head-up tilt, HDT rats demonstrated a lack of ability to maintain mean arterial pressure, similar to the orthostatic hypotension observed in astronauts upon return to Earth. It had been hypothesized that the orthostatic hypotension might be related to a diminished vasoconstrictor response in the splanchnic circulation. Few studies have been conducted to address this question. Therefore, a series of experiments were conducted to determine, 1) whether mesenteric veins and arteries exhibit diminished constrictor responsiveness, 2) the mechanism for the diminished vasoconstrictor responses, and 3) the possible stimulus for the hypo-responsiveness of the splanchnic vasculature. Results of these studies are in press in the Journal of Applied Physiology and the American Journal of Physiology: Regulatory, Integrative and Comparative Physiology. Because of the significance of the results in these studies, both are being published with an editorial to highlight the work. The results demonstrate that, within the same vascular bed, there is a reduced constriction of mesenteric arteries and veins to norepinephrine. This adrenergic hypo-responsiveness occurs in a vascular bed where blood flow and arterial pressure alterations do not occur as a result of the HDT intervention. Therefore, this suggests that the HDT-induced vascular effects on smooth muscle cells in this region are mediated by a systemic circulating factor(s) rather than altered hydrodynamic mechanical influences. Consistent with this hypothesis is an elevation of plasma atrial (ANP) and brain (NT-proBNP) natriuretic peptide concentrations in HDT rats, which were found to inhibit constrictor responses of mesenteric arteries and veins to adrenergic stimulation. The data demonstrate that in addition to norepinephrine, hindlimb unloading diminishes the vasoconstrictor responsiveness of mesenteric arteries to KCl and caffeine, but does not affect myogenic vasoconstriction within a physiological range of pressures. The reduced vasoconstrictor response to caffeine along with a lower RyR2 mRNA and protein expression indicates that the attenuated vasoconstrictor responsiveness of mesenteric arteries is the result of an impaired intracellular Ca2+ release. If similar alterations in mesenteric resistance artery function occur in humans during prolonged exposure to microgravity or bedrest, this would likely affect the ability to augment peripheral vascular resistance and cardiac output during orthostatic and exercise stresses. Thus, the functional consequences of these vascular alterations may include orthostatic and exercise intolerance commonly associated with actual and simulated microgravity.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
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